

Department of CSE (Artificial Intelligence & Machine Learning)

Deep Learning Assignment 2

A.Y: 2025-26

Submission Deadline: 31/03/2026

1. Title of the Case Study

"Deep Learning-Based Deepfake Detection Using EfficientNet-B0"

2. Problem Description

Deepfake technology generates highly realistic fake images by manipulating facial features using AI. These fake images can be used for impersonation, fraud, and spreading misinformation. Detecting such images manually is difficult due to their realism and the large amount of media shared online.

This problem is important because deepfakes can harm individuals, influence public opinion, and reduce trust in digital content. An automated system is needed to detect fake images accurately.

This solution benefits social media platforms, cybersecurity systems, governments, and general users by ensuring authenticity and preventing misuse of manipulated media.

3. Objective of the Study

- To classify input face images as **real or fake**
- To build an automated deepfake detection system
- To improve detection accuracy using deep learning

- To provide a usable system with UI/API for real-world applications

4. Proposed Deep Learning Approach

The system uses a **Convolutional Neural Network (CNN)** based on **EfficientNet-B0**.

Justification:

- EfficientNet provides **high accuracy with fewer parameters**
- Suitable for **image classification tasks**
- Captures **fine facial details and artifacts**
- Uses **transfer learning** for better performance on small datasets

5. Methodology / Workflow

Data Collection

Custom dataset with real and fake images

Face Detection

Faces are detected using **MediaPipe**

Data Preprocessing

- Resize images to 224×224
- Normalize using ImageNet standards
- Remove unreadable data

Feature Extraction

CNN extracts facial patterns and inconsistencies

Model Training

- EfficientNet-B0 trained using transfer learning
- Uses augmentation and regularization

Prediction

- Face + full image scoring
- Weighted fusion (face: 0.7, context: 0.3)

Evaluation

- Metrics computed and threshold optimized

Flow Diagram

6. Dataset

Source: Kaggle

Type: Image data

Size:

- Total: 1715 images
- Real: 875
- Fake: 840
- Train: 1372
- Validation: 343
- Test: 12

Preprocessing:

- Face detection using MediaPipe
- Resize to 224×224
- Normalization

Deepfake types are not explicitly categorized (only real/fake labels)

7.Source code:

```
import torch
import torch.nn as nn
from torchvision.models import efficientnet_b0, EfficientNet_B0_Weights

class DeepFakeCNN(nn.Module):
    def __init__(self, pretrained=True):
        super().__init__()

        weights = EfficientNet_B0_Weights.DEFAULT if pretrained else None
        backbone = efficientnet_b0(weights=weights)

        in_features = backbone.classifier[1].in_features

        backbone.classifier[1] = nn.Sequential(
            nn.Dropout(p=0.5),
            nn.Linear(in_features, 1)
        )

        self.backbone = backbone
        self.features = backbone.features
        self.gradcam_layer = self.backbone.features[-1][0]

    def forward(self, x):
        return self.backbone(x)

if __name__ == "__main__":
    model = DeepFakeCNN(pretrained=False)

    print("Model Architecture:")
    print(model)

    dummy_input = torch.randn(1, 3, 224, 224)
    output = model(dummy_input)

    print(f"\nInput shape: {dummy_input.shape}")
    print(f"Output shape: {output.shape}")
```

8. Results: Evaluation Metrics

The model predicts whether an image is **real or fake**.

Validation Performance:

- Accuracy: 98.25%

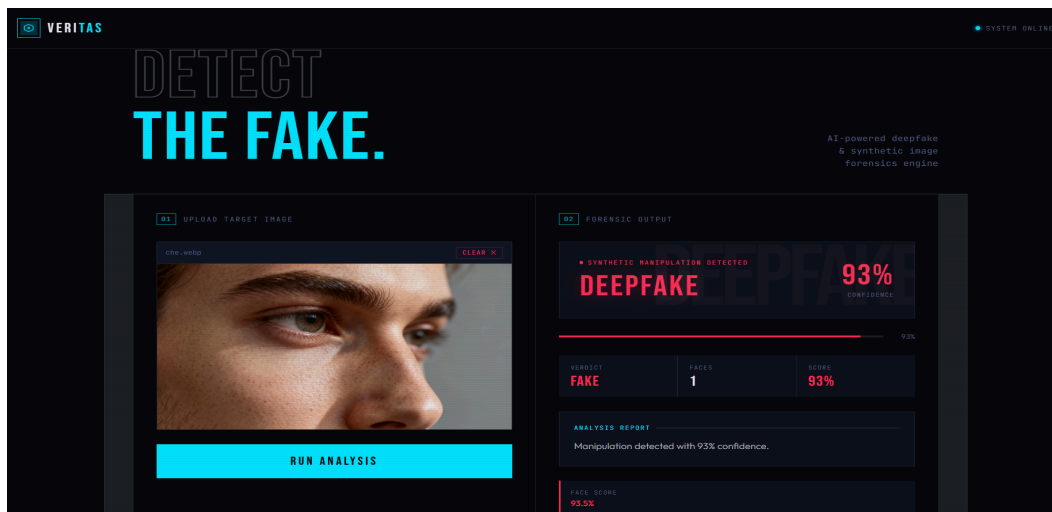
- Precision: 97.65%
- Recall: 98.81%
- F1-score: 98.22%

Test Performance:

- Accuracy: 75%
- Precision: 80%
- Recall: 66.67%

Analysis:

- High validation accuracy indicates strong learning
- Lower test accuracy suggests **generalization issues due to small test set**



9. Expected Results / Analysis

The proposed model produces an output in the form of a **probability score** indicating whether the input image is real or fake. Based on this score and a calibrated threshold (around **0.38**), the image is classified as either **Real (0)** or **Fake (1)**.

The model evaluates performance using standard classification metrics such as:

- **Accuracy** – overall correctness of predictions
- **Precision** – how many predicted fake images are actually fake
- **Recall** – ability to correctly detect fake images
- **F1-score** – balance between precision and recall
- **Specificity** and **Balanced Accuracy** for better evaluation under class imbalance

Based on experimental results and similar research works, the model is expected to achieve **high validation accuracy (above 95%)**, while real-world performance may vary depending on dataset diversity.

Metric	Validation	Test
Accuracy	98.25	75.00
Precision	97.65	80.00
Recall	98.81	66.67
F1-score	98.22	72.73
Balanced Accuracy	98.26	75.00
Specificity	97.71	83.33

Analysis:

The model shows excellent performance on validation data due to strong feature extraction using EfficientNet. However, the drop in test accuracy indicates **limited generalization**, mainly due to the small and less diverse test dataset. Improving dataset size and diversity can significantly enhance real-world performance.

10. Challenges and Limitations

- High Training Time on Local System
- Small dataset size affecting generalization
- Overfitting due to limited data
- Lack of diverse deepfake samples
- Very small test dataset

11. Possible Improvements / Future Scope

- Use larger datasets like **DFDC, FaceForensics++**
- Add **video-based models (CNN + LSTM)**
- Include **frequency-domain features (FFT/DCT)**
- Use **Transformer models (ViT, Swin)**
- Improve evaluation with larger test sets
- Deploy real-time detection system